6 **(a)** *α* : protons decrease by 2, neutron decrease by 2 [B1]

β : protons increase by 1, neutron decrease by 1 [B1]

γ: no change to number of protons and neutrons [B1]

(b) γ has no charge, so does not experience magnetic force [B1]. So its path should be undeflected [B1]

α and *β* have opposite charges, so should experience magnetic forces in opposite directions [B1]. So their paths should curve in opposite directions [B1]

(c) **(i)** Mass is converted [B1] into kinetic energies of nickel-60 nucleus and *β* particle (also energy of neutrino) as well as electromagnetic energy (photon energy) of the γ radiation [B1]

(ii) Energy released

$= (\text{mass loss})\ c^2$

$= (59.9338 - 59.9308 - 5.4348 \times 10^{-4})\ (1.66 \times 10^{-27})\ (3.00 \times 10^{8})^2$ [C1]

$= 3.67 \times 10^{-13}$ J [C1]

$= 2.29$ MeV [A1]

7 **(a)** **(i)** Sound cannot be polarised [B1] as it is longitudinal [B1]